7.2 Trigonometric Integrals

In Exercises 1–24, *find or evaluate the integral*.

1. $\int \sin^3 x\cos^2 x\,dx$ **2.** $\int_0^{\pi/2}\sqrt{\cos x}\,\sin^3 x\,dx$ **3.** $\int \cos^3 2x\,dx$

4. $\int \cos^4 x\,dx$ **5.** $\int \sin^2 2x\cos^4 2x\,dx$ **6.** $\int \cos^4 x\sin^4 x\,dx$

7. $\int \sin^6 u\,du$ **8.** $\int \theta\sin^2(\theta^2)\cos^2(\theta^2)\,d\theta$

9. $\int x\cos^2 x\,dx$ **Hint:** Integrate by parts. **10.** $\int \tan^3(\pi - x)\,dx$

11. $\int \tan^5 x\sec^3 x\,dx$ **12.** $\int_0^{\pi/4}\sec^2 x\tan^2 x\,dx$ **13.** $\int \sec^4(\pi - x)\tan(\pi - x)\,dx$

14. $\int \sec^4\left(\frac{x}{2}\right)\tan^4\left(\frac{x}{2}\right)dx$ **15.** $\int_{\pi/4}^{\pi/4}\cot^3 x\,dx$

16. $\int \csc^5 x\,dx$ **Hint:** Integrate by parts. **17.** $\int \csc^4\theta\cot^4\theta\,d\theta$

18. $\int \cot^3 x\csc^4 x\,dx$ **19.** $\int (1+\cot x)^2\csc x\,dx$ **20.** $\int \sin 3\theta\sin 4\theta\,d\theta$

21. $\int \frac{\sin^3 x}{\sec^2 x}dx$ **22.** $\int_0^{\pi/2}\frac{\sin t}{1+\cos t}dt$ **23.** $\int \frac{1}{\csc x\cot^2 x}dx$

24. $\int \frac{\cos 2\theta}{\cos\theta + \sin\theta}d\theta$

$\int \sin^3 x\cos^3 x\,dx$

$\int (1-\cos^2 x)\cos^3 x\sin x\,dx$ $u=\cos x$, $du=-\sin x\,dx$

$=\int (1-u^2)u^2(-du)$

$=\int u^4 - u^2\,du$

$=\frac{1}{5}u^5 - \frac{1}{3}u^3 + C$

$=\frac{1}{5}\cos^5 x - \frac{1}{3}\cos^3 x + C$

23. $\int \cancel{\sin x}\cdot\frac{\cos^2 x}{\sin^2 x}dx$ $u=\sin x$, $du=\cos x\,dx$

$=\int \frac{\cos^2 x}{\sin x}dx$ $\int du=\int\cos^2 x\,dx$

$v=\int\frac{1}{2}(1+\cos 2x)dx$

$=\frac{1}{2}\left[\int 1\,dx+\int\cos 2x\,dx\right]$

$=\frac{1}{2}\left[x+\frac{1}{2}\sin 2x\right]$

$=$

$=\sin x\cdot\frac{1}{2}\left[x+\frac{1}{2}\sin 2x\right]-\int\frac{1}{2}\left(x+\frac{1}{2}\sin 2x\right)\cos x\,dx$

25. Find the average value of $f(x) = \cos^2 x\sin^3 x$ over the interval $[0, \pi]$.

$=\frac{1}{2}x\sin x+\frac{1}{4}\sin x\sin 2x-\frac{1}{2}\left[\int x\cos x\,dx+\frac{1}{2}\int\sin 2x\cos x\,dx\right]$

26. Find the area of the region bounded by the graphs of $y = \sin^4 x$, $y = \cos^4 x$, $x = 0$, and $x = \pi/4$.

$=\frac{1}{2}x\sin x+\frac{1}{4}\sin x\sin 2x-\frac{1}{2}\left[x\sin x+\cos x+\frac{1}{2}\int 2\sin x\cos^2 x\,dx\right]$

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$=\cancel{\frac{1}{2}x\sin x}+\frac{1}{4}\sin x\cdot 2\sin x\cos x-\cancel{\frac{1}{2}x\sin x}-\frac{1}{2}\cos x-\frac{1}{2}\left(-\frac{1}{3}\cos^3 x\right)$

31. Prove that if m and n are positive integers, then

$$\int_{-\pi}^{\pi}\sin mx\cos nx\,dx = 0$$

$=\frac{1}{2}\sin^2 x\cos x-\frac{1}{2}\cos x+\frac{1}{6}\cos^3 x$

$=\frac{1}{2}\cdot\frac{1}{2}(1-\cos 2x)\cos x$

32. Prove that if m and n are positive integers, then

$$\int_{-\pi}^{\pi}\sin mx\sin nx\,dx = \begin{cases}0 & \text{if } m\neq n\\ \pi & \text{if } m = n\end{cases}$$

25. $\frac{1}{\pi-0}\int_0^{\pi} \cos^2 x \sin^3 x \, dx$

$= \frac{1}{\pi}\int_0^{\pi} \cos^2 x \sin^3 x \, dx$

$u = \cos 2x$

$du = 2\sin 2x \, dx$

$= 4\sin x \cos x \, dx$

$\frac{1}{4u} du = \sin x \, dx$

$\int \cos^2 x \sin^2 x \sin x \, dx$

$= \int \frac{1}{2}(1+\cos 2x)\cdot\frac{1}{2}(1-\cos 2x)\cdot \sin x \, dx$

$= \frac{1}{4}\int (1-\cos^2 2x)\sin x \, dx$

$= \frac{1}{4}\int (1-u^2)\frac{1}{4u} du$

$= \frac{1}{16}\int \frac{1-u^2}{u} du$

$= \frac{1}{16}\left(\int \frac{1}{u} du - \int u \, du\right)$

$= \frac{1}{16}\left(\ln|u| - \frac{1}{2}u^2\right)$

$= \frac{1}{16}\left(\ln|\cos 2x| - \frac{1}{2}\cos^2 2x\right)$

$\frac{1}{\pi} \cdot \left(\frac{1}{16}\left(\ln|\cos 2x| - \frac{1}{2}\cos^2 2x\right)\Big|_0^{\pi}\right)$

$= \frac{1}{\pi}\left[\frac{1}{16}\left(\cancel{\ln 1}^{0} - \frac{1}{2}\cdot 1\right) - \frac{1}{16}\left(\cancel{\ln 1}^{0} - \frac{1}{2}\cdot 1\right)\right] = 0$ #

33. Prove that if m and n are positive integers, then

$$\int_{-\pi}^{\pi} \cos mx \cos nx\,dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}$$

34. Finite Fourier Series Prove that if

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{n} (a_k \cos kx + b_k \sin kx)$$

$$= \frac{a_0}{2} + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots + a_n \cos nx + b_n \sin nx$$

then

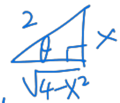
$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx, \qquad a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos kx\,dx$$

and

$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin kx\,dx \qquad k = 1, 2, \ldots, n$$

Hint: Use the results of Exercises 31-33

1. $\int \frac{\sqrt{4-x^2}}{x^2}\,dx$ $x = 2\sin\theta$, $dx = 2\cos\theta\,d\theta$ $u = \sin^2\theta$, $du = 2\sin\theta\cos\theta\,d\theta$ $\frac{x}{2} = \sin\theta$



$= \int \frac{\cancel{2}\cos\theta}{\cancel{4}\sin^2\theta}\cdot\cancel{2}\cos\theta\,d\theta = \int \frac{\cos^2\theta}{\sin^2\theta}\,d\theta = \int \cot^2\theta\,d\theta = \int(\csc^2\theta - 1)\,d\theta = \int \csc^2\theta\,d\theta - \int 1\,d\theta$

$= -\cot\theta - \theta + C$

$= \frac{-\sqrt{4-x^2}}{x} - \sin^{-1}\left(\frac{x}{2}\right) + C$ #

7.3 Trigonometric Substitutions

In Exercises 1–16, *find or evaluate the integral using an appropriate trigonometric substitution.*

1. $\int \frac{\sqrt{4-x^2}}{x^2}dx$

2. $\int \frac{1}{x^2\sqrt{1-x^2}}dx$

3. $\int x^3\sqrt{1+x^2}\,dx$

4. $\int \frac{1}{x^3\sqrt{x^2-4}}dx$

5. $\int x^3\sqrt{4-x^2}\,dx$

6. $\int_0^{3/4} \frac{x^2}{\sqrt{9-4x^2}}dx$

7. $\int (4-x^2)^{3/2}dx$

8. $\int \frac{x^2}{\sqrt{3-x^2}}dx$

9. $\int \frac{\sqrt{9x^2+4}}{x^4}dx$

10. $\int \frac{1}{(9+x^2)^2}dx$

11. $\int_2^4 \frac{\sqrt{x^2-4}}{x^4}dx$

12. $\int \frac{2x+3}{\sqrt{1-x^2}}dx$

13. $\int e^t\sqrt{1+e^{2t}}\,dt$

14. $\int \sqrt{4x-x^2}\,dx$

15. $\int \frac{t^2}{\sqrt{4t-t^2}}dt$

16. $\int \frac{1}{(3-2x-x^2)^{5/2}}dx$

23. Evaluate

$$\int_0^{\pi/4} \frac{dx}{a^2\cos^2 x + b^2\sin^2 x} \qquad a>0, \quad b>0$$

Hint: Use the substitution $u = \tan x$

7. $\int (4-x^2)^{\frac{3}{2}}\,dx$ $\frac{x}{2} = \sin\theta$, $x = 2\sin\theta$, $dx = 2\cos\theta\,d\theta$

$\int \left(\sqrt{4-4\sin^2\theta}\right)^3 \cdot 2\cos\theta\,d\theta$

$= \int (2\cos\theta)^4\,d\theta$

$= 16\int \cos^4\theta\,d\theta$

$= 16\int \left[\frac{1}{2}(1+\cos 2\theta)\right]^2 d\theta$

$= 4\int 1 + 2\cos 2\theta + \cos^2 2\theta\,d\theta$

$= 4\left(\int 1\,d\theta + 2\int \cos 2\theta\,d\theta + \int \cos^2 2\theta\,d\theta\right)$

$= 4\left(\theta + \cancel{2}\cdot\frac{1}{\cancel{2}}\sin 2\theta + \frac{1}{2}\theta + \frac{1}{8}\sin 4\theta\right)$

$= 6\theta + 4\sin 2\theta + \frac{1}{2}\sin 4\theta$

$6\sin^{-1}\left(\frac{x}{2}\right) + 4\cdot 2\sin\theta\cos\theta + \frac{1}{\cancel{2}}\cdot\cancel{2}\,\cancel{\sin 2\theta}\cdot\cancel{\cos 2\theta}$ $2\sin\theta\cos\theta$ $(\cos^2\theta - \sin^2\theta)$

$6\sin^{-1}\left(\frac{x}{2}\right) + 8\cdot\sin\theta\cos\theta + 2\sin\theta\cos\theta(\cos^2\theta - \sin^2\theta)$

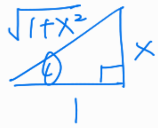
$= 6\sin^{-1}\left(\frac{x}{2}\right) + \cancel{8}^{4}\cdot\frac{x}{\cancel{2}}\cdot\frac{\sqrt{4-x^2}}{\cancel{2}} + \cancel{2}\cdot\frac{x}{\cancel{2}}\cdot\frac{\sqrt{4-x^2}}{2}\cdot\left(\frac{4-x^2}{4} - \frac{x^2}{4}\right)$

3. $\int x^3\sqrt{1+x^2}\,dx$

$x = \tan\theta$

$dx = \sec^2\theta\,d\theta$

$\tan^2\theta + 1 = \sec^2\theta$



$= \int \tan^3\theta\cdot\sec\theta\cdot\sec^2\theta\,d\theta$

$u = \sec\theta$

$du = \sec\theta\tan\theta\,d\theta$

$= \int \tan^2\theta\cdot\sec^2\theta\cdot\sec\theta\tan\theta\,d\theta$

$= \int (u^2-1)\,u^2\cdot du$

$= \int (u^4-u^2)\,du = \int u^4\,du - \int u^2\,du$

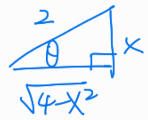
$= \frac{1}{5}u^5 - \frac{1}{3}u^3 + C = \frac{1}{5}\sec^5\theta - \frac{1}{3}\sec^3\theta + C$

$= \frac{1}{5}\left(\sqrt{1+x^2}\right)^5 - \frac{1}{3}\left(\sqrt{1+x^2}\right)^3 + C$ #

5 $\int x^3\sqrt{4-x^2}\,dx$

$x = 2\sin\theta$

$dx = 2\cos\theta\,d\theta$



$= \int 8\sin^3\theta\,\underbrace{\cancel{\sqrt{4-4\sin^2\theta}}}^{2\cos\theta}\cdot 2\cos\theta\,d\theta$

$u = \cos\theta$

$du = -\sin\theta\,d\theta$

$= 32\int \sin^3\theta\cos^2\theta\,d\theta$

$= 32\int \sin^2\theta\cos^2\theta\cdot\sin\theta\,d\theta$

$= 32\int (1-u^2)\,u^2(-du)$

$= 32\int u^4 - u^2\,du$

$= 32\left(\frac{1}{5}u^5 - \frac{1}{3}u^3\right) + C$

$= 32\left(\frac{1}{5}\cos^5\theta - \frac{1}{3}\cos^3\theta\right) + C$

$= \frac{32}{5}\cdot\left(\frac{\sqrt{4-x^2}}{2}\right)^5 - \frac{32}{3}\left(\frac{\sqrt{4-x^2}}{2}\right)^3 + C$ #

7.4 The Method of Partial Fractions

In Exercises 1–3 *, write the form of the partial fraction decomposition of the rational expression. Do not find the numerical values of the constants.*

1. a. $\dfrac{2x+1}{x^2-x-2}$ **b.** $\dfrac{x-4}{x^2+4x+3}$

2. a. $\dfrac{2x+1}{x^3+x}$ **b.** $\dfrac{8x}{x^3-5x^2}$

3. a. $\dfrac{2x^3-3x-5}{x^2(x+1)^3}$ **b.** $\dfrac{2x^4-3x^2+8x+1}{(x-1)^2(x^2+x+1)^3}$

In Exercises 4–25*, find or evaluate the integral.*

4. $\displaystyle\int \frac{3x+2}{x(x-2)}dx$ **5.** $\displaystyle\int \frac{2x-1}{2x^2-x}dx$ **6.** $\displaystyle\int \frac{1}{4x^2-9}dx$

7. $\displaystyle\int_0^1 \frac{2u+3}{u^2+4u+3}du$ **8.** $\displaystyle\int \frac{x^2+2x+8}{x^3-4x}dx$ **9.** $\displaystyle\int_2^3 \frac{x^3-2x+7}{x^2+x-2}dx$

10. $\displaystyle\int \frac{x^4-3x^2-3x-2}{x^3-x^2-2x}dx$ **11.** $\displaystyle\int_2^4 \frac{3x-5}{(x-1)^2}dx$ **12.** $\displaystyle\int_1^2 \frac{x^2+10x-36}{x(x-3)^2}dx$

13. $\displaystyle\int \frac{4x^2}{(x^2-4)^2}dx$ **14.** $\displaystyle\int \frac{x^2}{(x^2+4x+3)^2}dx$ **15.** $\displaystyle\int \frac{x^2+16x+7}{x^3-x^2+x+3}dx$

16. $\displaystyle\int \frac{2r^2-3r+4}{(r^2+2)^2}dr$ **17.** $\displaystyle\int \frac{13x+4}{(x-2)(x^2+2x+2)}dx$ **18.** $\displaystyle\int \frac{x^2+1}{x^3-1}dx$

19. $\displaystyle\int \frac{x^2-x-21}{2x^3-x^2+8x-4}dx$ **20.** $\displaystyle\int \frac{3x-x^2}{(x^2+1)(x^2+2)}dx$ **21.** $\displaystyle\int \frac{t^4}{t^4-1}dt$

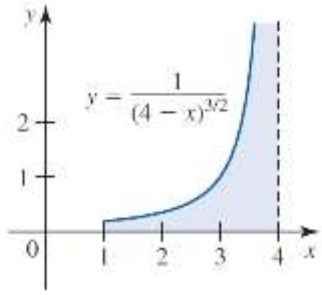
22. $\displaystyle\int \frac{\cos x}{\sin^2 x-\sin x-6}dx$ **23.** $\displaystyle\int \frac{\sec^2\theta}{\tan\theta(\tan\theta-1)}d\theta$ **24.** $\displaystyle\int \frac{e^x}{e^{2x}+2e^x-8}dx$

25. $\displaystyle\int \frac{dx}{e^x(1+e^{2x})}dx$

7.6 Improper Integrals

In Exercises 1–3, *find the area of the shaded region, if it exists*.

1.



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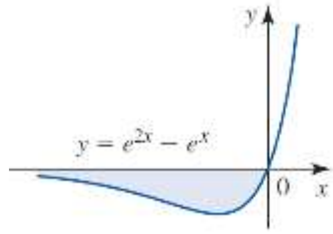
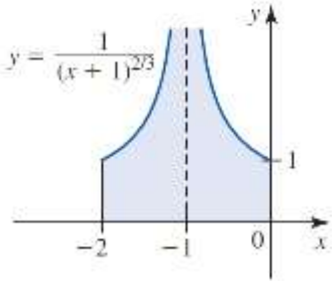


3.



In Exercises 4–21, *determine whether the improper integral converges or diverges, and if it converges, find its value*.

4. $\displaystyle\int_1^{\infty} \frac{1}{x^{0.99}}\,dx$ **5.** $\displaystyle\int_0^{\infty} \frac{1}{(x+1)^2}\,dx$ **6.** $\displaystyle\int_2^{\infty} \frac{1}{\sqrt[3]{x-1}}\,dx$

7. $\displaystyle\int_e^{\infty} \frac{1}{x\ln^2 x}\,dx$ **8.** $\displaystyle\int_0^{\infty} e^{-x}\sin x\,dx$ **9.** $\displaystyle\int_{-\infty}^{0} \frac{1}{x^2+2x+5}\,dx$

10. $\displaystyle\int_{-\infty}^{\infty} xe^{-x^2}\,dx$ **11.** $\displaystyle\int_{-\infty}^{\infty} \cos^2 x\,dx$ **12.** $\displaystyle\int_{-\infty}^{\infty} e^{-|x|}\,dx$

13. $\displaystyle\int_{-2}^{1} \frac{1}{x^2}\,dx$ **14.** $\displaystyle\int_0^{2} \frac{1}{2x-3}\,dx$ **15.** $\displaystyle\int_0^{2} \frac{1}{x^2-2x}\,dx$

16. $\displaystyle\int_0^{e} \ln x\,dx$ **17.** $\displaystyle\int_0^{\pi} \sec^2 x\,dx$ **18.** $\displaystyle\int_0^{\pi/2} \tan^2 x\,dx$

19. $\displaystyle\int_0^{\infty} \frac{\sqrt{\tan^{-1} x}}{1+x^2}\,dx$ **20.** $\displaystyle\int_0^{\infty} \frac{1}{e^x-1}\,dx$ **21.** $\displaystyle\int_1^{\infty} \frac{dx}{x\sqrt{x^2-1}}$

In Exercises 22– 24, *use the Comparison Test to determine whether the integral is convergent or divergent by comparing it with the second integral.*

22. $\displaystyle\int_1^{\infty} \frac{1}{\sqrt{x^3+1}}\,dx;\quad \int_1^{\infty} \frac{1}{x^{3/2}}\,dx$ **23.** $\displaystyle\int_1^{\infty} \frac{dx}{x+\sin^2 x};\quad \int_1^{\infty} \frac{1}{1+x}\,dx$

24. $\displaystyle\int_1^{\infty} \frac{1}{\sqrt{1+x^2+x^4}}\,dx;\quad \int_1^{\infty} \frac{1}{x^2}\,dx$